ROLLUPOS APPLICATION USER MANUAL • AUTOMOTIVE & TELEMETRY

⚡ Blue Oval Garage

Blue Oval Garage v2.0 — 2011–2026+ Ford & Lincoln Multi-Vehicle FORScan Companion, VIN Barcode/OCR Scanner & Deterministic As-Built Engine

Version: v2.0.0 | Build Hash: #a36e508a | 🔒 Per-User Authorized Cartridge | Auto-Synced: 2026-10-26 22:50

1. ⚡ 60-Second Quickstart Guide

1 **Select or Add Any 2011–2026+ Ford/Lincoln (`🔵 Garage & VIN`)**
Your 2023 F-150 Lightning ER Lariat (`1FT6W1EV4PWG11434`, ⚡ BEV) is pre-seeded as Vehicle #1. Tap `+ Add Vehicle` to add any 2011–2026+ Ford or Lincoln by typing a 17-char VIN or scanning the door-jamb B-pillar sticker / windshield VIN plate with the Hybrid Camera Barcode + OCR Scanner.

2 **Load a Baseline (`.abt` / `.ab` or `💉 Demo Baseline`)**
Tap `📂 Import .abt / .ab File` to load your real FORScan `.abt` module backups or Ford Motorcraft `.ab` XML—or tap `💉 Load Demo Baseline` to preview live nibble mask and checksum dry-runs immediately for the active vehicle.

3 **Browse 183 Recipes, 21 Master Spreadsheets & Custom Lines (`🎛️ Mod Menu`)**
Browse 183 verified recipes across 6 upstream platforms (`f150`, `super_duty`, `bronco`, `explorer`, `ranger_6g`, `mustang`), download 1-click `.xlsx`/`.pdf` community master spreadsheets (Livnitup, ZooDad, DanMc85) for all 21 platforms, or use `➕ Quick-Add Custom Spreadsheet Line` to enter any forum/spreadsheet hex mod with deterministic checksum dry-run.

4 **Run Your In-Cab FORScan Session (`📋 Session Plan`)**
Open the `📋 Plan` tab in the vehicle cab with your OBDLink EX (`STP31` HS-CAN + automatic `STP33` 125 kbaud MS-CAN guidance for 2011–2020 CGEA 1.2/1.3 vehicles). Verify every target module has a saved `.abt` backup and copy/share your exact line-by-line write sheet.

2. 🧭 Screen-by-Screen & Navigation Map

SCREEN / TAB / MODAL	WHAT IT CONTROLS & HOW TO USE IT
🔵 **Garage & VIN**	Multi-vehicle garage switcher, Hybrid Camera Barcode/OCR VIN scanner with ISO 3779 Modulo-11 auto-repair, offline + NHTSA vPIC decoder, and JSON Vault export/import.
🎛️ **Mod Menu (183 Recipes + Custom)**	3-Layer FORScan catalog: 183 platform-filtered recipes, 21 curated community master spreadsheets (.xlsx/.pdf), and Quick-Add Custom Spreadsheet Line builder.
🧩 **Modules (87 Addresses)**	Complete Ford/Lincoln control module directory (`726`, `7D0`, `720`, `740/741`, `7E7`, `746`, etc.) with CGEA 1.2/1.3 MS-CAN (`STP33`) notes, write-risk ratings, and per-vehicle backup evidence.
🛡️ **Backups & Factory .ab**	Per-vehicle module backup coverage checker, `.abt`/`.ab` importer, Factory vs. Current byte drift differ, Motorcraft AsBuilt helper, and Checksum Calculator.
📋 **Session Plan**	Printable/shareable in-cab FORScan write sheet with CGEA 1.2/1.3 vs. FNV2 adapter guidance, required module backups, and exact target hex lines.

3. 🔬 In-Depth Feature Guide & Workflows

Hybrid VIN Barcode + Optical Confusable Auto-Repair (`Code -> Vision -> Code`)

Combines native `BarcodeDetector` (Code 39 / DataMatrix door-jamb stickers with automatic 18th-char `'I'` guard stripping), Canvas CLAHE contrast enhancement, and a deterministic ISO 3779 Position 9 Modulo-11 repair engine that tests single- and double-character optical confusable substitutions (`O/Q/D -> 0`, `I/L -> 1`, `Z -> 2`, `S -> 5`, `G -> 6`, `B -> 8`) until the mathematical check digit passes.

Full 3-Layer Vehicle-Specific FORScan Catalog & Custom Mod Builder

Layer 1 ships 183 structured recipes across F-150/Lightning, Super Duty, Bronco, Explorer/Aviator, Ranger 6G, and Mustang with dynamic powertrain gating (`ICE_ONLY` blocked on `BEV`, unlocked on `ICE_GAS`/`ICE_DIESEL`/`HEV`). Layer 2 provides 1-click `.xlsx`/`.pdf` community spreadsheets for 21 Ford/Lincoln platforms (2011–2026+). Layer 3 lets you add any custom spreadsheet line (`726-12-01`, `xxxx-**xx-xxxx`) with full nibble-mask and Modulo-256 checksum dry-runs.

100% Deterministic Nibble Mask, Checksum & Multi-Vehicle Isolation

Evaluates `set`, `bits`, and `offset` nibble operations across multi-step recipes and recomputes exact Ford line checksums (`sum(header + data[:-1]) & 0xFF`). Every vehicle in your garage maintains isolated `.abt`/`.ab` backups, starred mods, custom lines, and session verification history.

4. ⚙️ Settings, Secret Vault & Multi-User Data Isolation

SETTING KEY	FIELD DESCRIPTION	TYPE	DEFAULT
`default_vin`	Default 17-Character Vehicle VIN	`text`	`1FT6W1EV4PWG11434`
`default_trim`	Default Trim Level	`text`	`Lariat`
`default_battery`	Default Battery Pack or Engine	`text`	`Extended Range (131 kWh)`

🔒 **Storage Namespace Pattern:** `rollup::v1::u::<google-email>::app::lightning-garage::*`

📦 **Active App Storage Keys:** `__user_config__` (Managed automatically)

🔐 **Private On-Device Vault Secrets:** None required (uses Hub session)

Isolated Per-User Storage (RollupBridge) | Mobile Camera / Multimodal Capture | Android Share Sheet & CSV/JSON Export | Encrypted On-Device Secret Vault

5. 🎛️ Live UI Controls Reference & Recent Updates

Interactive Controls Detected in Current Build (`v2.0.0`):

📖 Guide | 🚗 Garage & VIN | 🎛️ Mod Menu (183) | 🧩 Modules (87) | 🛡️ Backups & Factory .ab | 📋 Session Plan (0)

🚗 Garage/VIN | 🎛️ Mod Menu | 🧩 Modules | 🛡️ Backups/.ab | 📋 Plan (0)

Pro Tips & Best Practices:

- When scanning a door-jamb barcode on Ford trucks, the deterministic normalizer automatically strips the leading `'I'` Code 39 guard character from 18-character barcode payloads.
- For 2011–2020 CGEA 1.2/1.3 vehicles, modules `740 DDM`, `741 PDM`, `733 HVAC`, and `791 TRM` reside on the 125 kbaud MS-CAN bus—use an electronic bus-switching cable like OBDLink EX (`STP33`).

Recent Cartridge Update Log (Auto-Tracked on Publish):

- `61ab551` (2026-10-26) — RollupOS Hub & Mini-App sync (2026-10-26 22:40)
- `2721026` (2026-10-26) — feat(gravity): sync HTTPS tunnel & mobile UX to GitHub Pages (2026-10-26 16:41)
- `14a7159` (2026-10-26) — feat(gravity): sync HTTPS tunnel & mobile UX to GitHub Pages (2026-10-26 16:40)
- `c4e5aad` (2026-10-26) — feat(gravity): sync HTTPS tunnel & mobile UX to GitHub Pages (2026-10-26 16:38)